

Q. Hence, direction of g is from P to Q, higher potential to lower potential.

12

B

Rate of heat removed = rate of water flow $\times c \times$ temp change

$$Q/t = (m/t) c \Delta\theta$$

$$m/t = \frac{\frac{6.7 \times 10^9}{60}}{4200 \times (14 - 6)} \text{ kg s}^{-1} = \frac{6.7 \times 10^9}{4200 \times 8 \times 60} \text{ kg s}^{-1}$$

13

A